The core of G5

Motegi's all-cardinality branching matrix is its own transpose-complement inverse: a complete proof of the type-free case, and hence of both types

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21 October 2026

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Commit hash (PROTOCOL 2.3). `clio-vega/proofs` commit `97158d7` — the commit that introduced this `.tex` and its PDF (written against `HEAD` = `421ae34`; the hash above was filled in by the immediately following commit, which is the only way a file can name the commit that introduces it, and both are therefore machine-resolvable); repository `https://github.com/clio-vega/proofs`. The repository is `~/projects/proofs`; `~/projects` is *not* a repository and `git rev-parse` run there emits a `fatal:` that has once been transcribed into a PDF as "no commit hash exists". Read-only companion repo `~/git/puzzles` at `9b30b8f`.

Primary source. K. Motegi, *Type B- and C- trigonometric weight functions*, `arXiv:2610.15344`, LaTeX source `typebcweightfunctions_arxivversionfinal-3.tex`, 6430 lines, `md5 c15a23f2f94f4f539f67ba21d5e2f864` — the *content hash*, which is the only cache stamp that distinguishes a truncation from an edit. All line numbers `L`n refer to that file.

Abstract

Motegi's all-cardinality biorthogonality (`thm:BC-all-cardinality-biorthogonality`, L4365; our "G5") rests on `prop:allcard-Bhat-inverse` (L4897), whose published proof turns on an unproved bookkeeping claim: that every two-index factor of two expanded branching coefficients is accounted for exactly once. This note proves that claim and assembles a complete, gap-free proof of the proposition in the type-free normalisation $c \equiv 1$; by Theorem A of [C20] this gives both types and all N. Two results, not one. *First* (§2) — and this gap was not on the agenda — `lem:allcard-Bhat-product` (L4660), the closed form for the normalised branching coefficient, is itself proved only by the words "substituting these four expressions and simplifying". I give the simplification as seven exact cancellations, for both types at once, and in doing so locate the entire type-B/type-C difference: it is the *diagonal* $a = b$ term of a pair product in type C against a genuine single-index product in type B. *Second* (§3–§6), the two-index accounting is a bijection — but "bijection" is one word covering two claims with *different* load-bearing structure: surjectivity needs only the set partition $[N] = \mathbf{L} \sqcup \mathcal{D} \sqcup \mathbf{M}^{\mathrm{c}}$, while injectivity needs the *total order* on $[N]$, and fails without it. Two pre-registered lesions confirm the split by breaking disjoint halves. What this does *not* do is close G5 outright; §8 states exactly which two premises remain and at what trust — and reports a third thing found while writing that section: *the entire type-B branch of [Mot] is stated without proof* (L3598, "parallel … and are therefore omitted"), so every type-B statement upstream of this note is a declared parallelism rather than a demonstration. Lemma 3 discharges that parallelism for the one step this note needs, and leaves the rest open.

1 What was open

Let $F = \mathbb{Q}(q, w_1, \dots, w_N)$, let $\mathbf{A}^{\mathrm{c}} = [N] \setminus \mathbf{A}$ and $|\lambda(\mathbf{A})| = \sum_{a \in \mathbf{A}} a$. Motegi factors the matrix of specialised type-B/C weight functions as $\mathsf{A}^X = \widehat{\mathrm{Br}}^X \widehat{\Phi}$ (`eq:allcard-P-factorization`, L4358), where $\widehat{\Phi}$ is block diagonal by cardinality and carries all the type-A content, and $\widehat{\mathrm{Br}}^X$ is unitriangular for inclusion and carries all the type dependence. G5 then needs the inverse of each factor. The $\widehat{\Phi}$ half is `lem:allcard-Phihat-inverse` and imports Rimányi–Tarasov–Varchenko. The $\widehat{\mathrm{Br}}^X$ half is

`prop:allcard-Bhat-inverse` (L4897). For $X \in \{B, C\}$, $\big(\widehat{\mathrm{Br}}^X(\mathbf{w}, q)^{-1}\big)_{\mathbf{I},\mathbf{M}} = q^{|\lambda(\mathbf{I})| - |\lambda(\mathbf{M})|}\, \widehat{\mathrm{Br}}^X_{\mathbf{M}^{\mathrm{c}},\mathbf{I}^{\mathrm{c}}}(\mathbf{w}, q^{-1})$,

and needs no type-A input anywhere. I traced the chain on 2026-10-20 and audited this branch; the audit found no error, but it left one step with no proof in it. After expanding both branching coefficients (L5093 and L5126) and organising their factors, Motegi writes (L5263):

> *"Every two-index factor in (L5093) and (L5126) has now been included either in some $E_j^{(\epsilon_j)}$ or in some $\mathsf{P}_{\epsilon_i\epsilon_j}(x_i, x_j)$."*

That sentence is the hinge of the proof and it is an assertion. It is a finite combinatorial accounting statement over pairs, and it is what §3 proves. A second, unadvertised step of the same shape sits one lemma earlier and is dealt with first.

Throughout, the object is the type-free kernel of [C20].

Definition 1 (the type-free kernel; `def:G` of [C20])**.** For $\mathbf{K} \subseteq \mathbf{M} \subseteq [N]$ put $\Omega := \mathbf{M} \setminus \mathbf{K}$ and

$$G_{\mathbf{M},\mathbf{K}}(\mathbf{w}, q) := (1-q)^{|\Omega|} \prod_{\substack{a<b \\ a,b \in \Omega}} \frac{1 - q w_a^{-1} w_b^{-1}}{1 - w_a^{-1} w_b^{-1}} \times \prod_{\substack{k \in \mathbf{K},\ t \in \Omega \\ k<t}} \frac{1 - q w_k^{-1} w_t}{1 - w_k^{-1} w_t} \prod_{\substack{t \in \Omega,\ j \in \mathbf{M}^{\mathrm{c}} \\ j<t}} \frac{1 - q w_t^{-1} w_j}{1 - w_t^{-1} w_j}, \tag{1}$$

and $G_{\mathbf{M},\mathbf{K}} := 0$ if $\mathbf{K} \not\subseteq \mathbf{M}$. Every denominator in (1) is a nonzero element of F, so $G_{\mathbf{M},\mathbf{K}} \in F^{\times}$; all identities below are identities of rational functions and no substitution of a value for q or w_i is ever made.

Definition 2. For $c = (c_1, \dots, c_N)$ with $c_t \in F^{\times}$ put $B^c_{\mathbf{M},\mathbf{K}} := \big(\prod_{t \in \mathbf{M} \setminus \mathbf{K}} c_t\big) G_{\mathbf{M},\mathbf{K}}$ for $\mathbf{K} \subseteq \mathbf{M}$ and 0 otherwise, and

$$T^c_{\mathbf{M},\mathbf{L}} := \sum_{\mathbf{K} \subseteq [N]} B^c_{\mathbf{M},\mathbf{K}}(\mathbf{w}, q)\; q^{|\lambda(\mathbf{K})| - |\lambda(\mathbf{L})|}\; B^c_{\mathbf{L}^{\mathrm{c}},\mathbf{K}^{\mathrm{c}}}(\mathbf{w}, q^{-1}). \tag{2}$$

Write $\mathbf{1} = (1, \dots, 1)$, so $B^{\mathbf{1}} = G$.

The target of this note is

$$T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}} = \delta_{\mathbf{M},\mathbf{L}} \qquad \text{for all } \mathbf{M}, \mathbf{L} \subseteq [N]. \tag{3}$$

Theorem A of [C20] states that for any c with $c_t(\mathbf{w}, q^{-1}) = c_t(\mathbf{w}, q)$ one has $T^c_{\mathbf{M},\mathbf{L}} = \big(\prod_{d \in \mathbf{M} \setminus \mathbf{L}} c_d\big) T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}}$ when $\mathbf{L} \subsetneq \mathbf{M}$ and $T^c = T^{\mathbf{1}}$ otherwise, so that (3) holds for one such c if and only if it holds for every one. Hence (3) gives `prop:allcard-Bhat-inverse` in both types at once, once the two types are identified as two values of c — which is the next section.

2 The unadvertised gap: the closed form of $\widehat{\mathrm{Br}}^X$

`lem:allcard-Bhat-product` (L4660) asserts $\widehat{\mathrm{Br}}^X_{\mathbf{M},\mathbf{K}} = \big(\prod_{t\in\Omega} c_t^X\big) G_{\mathbf{M},\mathbf{K}}$ with

$$c_t^B = \frac{1}{1-w_t^{-1}}, \qquad c_t^C = \frac{1}{1-w_t^{-2}}.$$

Its published proof lists the four ingredients and then says: "Substituting these four expressions and simplifying, using $\mathbf{M} = \mathbf{K} \sqcup \Omega$, $\mathbf{K}^{\mathrm{c}} = \Omega \sqcup \mathbf{M}^{\mathrm{c}}$, gives `eq:allcard-Bhat-product-C`. The type B formula is obtained by the same simplification." In [C20] I recorded this as verified in 240/240 instances and therefore `computed`; a census is not a derivation. Here is the derivation.

Lemma 3. *Let $\mathbf{K} \subseteq \mathbf{M} \subseteq [N]$, $\Omega = \mathbf{M} \setminus \mathbf{K}$, $X \in \{B, C\}$. With $\widehat{\mathrm{Br}}^X$ as in `eq:allcard-Bhat-def` (L4312), Br^X as in `eq:allcard-Br-C/-B` (L4212/L4233), D^X as in `eq:local-DC-orth/-DB-orth` (L3846/L3858) and the diagonal $\Phi_{\mathbf{A},\mathbf{A}}$ as in `RTVtheorem`(ii) (L4745ff),*

$$\widehat{\mathrm{Br}}^X_{\mathbf{M},\mathbf{K}} = \frac{\mathrm{Br}^X_{\mathbf{M},\mathbf{K}}(\mathbf{w},q)\,\Phi_{\mathbf{K},\mathbf{K}}(\mathbf{w},q)}{D^X_{|\mathbf{M}|}(\mathbf{M};\mathbf{w},q)\,\Phi_{\mathbf{M},\mathbf{M}}(\mathbf{w},q)} = \Big(\prod_{t\in\Omega} c_t^X\Big)\, G_{\mathbf{M},\mathbf{K}}(\mathbf{w},q).$$

Proof. Write the four ingredients out. For type C,

$$\mathrm{Br}^C_{\mathbf{M},\mathbf{K}} = (1-q)^{|\Omega|} \underbrace{\prod_{\substack{a\le b\\ a,b\in\mathbf{M}^{\mathrm{c}}}} (1-qw_aw_b)}_{(\alpha)} \underbrace{\prod_{\substack{a\le b\\ a,b\in\mathbf{K}}} (1-w_a^{-1}w_b^{-1})}_{(\beta)} \underbrace{\prod_{\substack{k\in\mathbf{K}\\ t\in\Omega}} \frac{1-w_k^{-1}w_t^{-1}}{1-w_k^{-1}w_t}}_{(\gamma)}$$
$$\times \underbrace{\prod_{\substack{a<b\\ a,b\in\Omega}} (1-qw_a^{-1}w_b^{-1})}_{(\delta)} \underbrace{\prod_{\substack{t\in\Omega\\ j\in\mathbf{M}^{\mathrm{c}}}} (1-qw_t^{-1}w_j)}_{(\zeta)},$$

$$D^C_{|\mathbf{M}|}(\mathbf{M}) = \prod_{\substack{a\le b\\ a,b\in\mathbf{M}}} (1-w_a^{-1}w_b^{-1}) \cdot (\alpha),$$

and, with $\mathbf{A}^{\mathrm{c}}$ the complement of $\mathbf{A}$,

$$\Phi_{\mathbf{A},\mathbf{A}} = \prod_{\substack{a\in\mathbf{A},\ u\in\mathbf{A}^{\mathrm{c}}\\ u<a}} (1-w_a^{-1}w_u) \prod_{\substack{a\in\mathbf{A},\ u\in\mathbf{A}^{\mathrm{c}}\\ a<u}} (1-qw_a^{-1}w_u) \qquad (\mathbf{A}\in\{\mathbf{K},\mathbf{M}\}).$$

Now seven cancellations.

(1) The factor (α) occurs identically in $\mathrm{Br}^C_{\mathbf{M},\mathbf{K}}$ and in $D^C_{|\mathbf{M}|}(\mathbf{M})$ and cancels. (This is the whole of the $\mathbf{M}^{\mathrm{c}}$-only content: it depends on $\mathbf{M}$ but not on $\mathbf{K}$, so it must cancel in a matrix normalised by its own diagonal.)

(2) Since $\mathbf{M} = \mathbf{K} \sqcup \Omega$ and $(1-w_a^{-1}w_b^{-1})$ is symmetric in a, b, the pairs $a \le b$ in $\mathbf{M}$ split into those inside $\mathbf{K}$, those inside Ω, and the mixed ones, each mixed unordered pair occurring once:

$$\prod_{\substack{a\le b\\ a,b\in\mathbf{M}}} (1-w_a^{-1}w_b^{-1}) = (\beta) \cdot \prod_{\substack{a\le b\\ a,b\in\Omega}} (1-w_a^{-1}w_b^{-1}) \cdot \prod_{\substack{k\in\mathbf{K}\\ t\in\Omega}} (1-w_k^{-1}w_t^{-1}).$$

So (β) cancels and leaves the last two products in the denominator.

(3) The denominator factor $\prod_{k\in\mathbf{K},t\in\Omega}(1-w_k^{-1}w_t^{-1})$ produced by (2) is exactly the numerator of (γ). They cancel, leaving $\prod_{k\in\mathbf{K},t\in\Omega}(1-w_k^{-1}w_t)^{-1}$.

(4) The remaining denominator factor from (2) splits off its diagonal:

$$\prod_{\substack{a\le b\\ a,b\in\Omega}}(1-w_a^{-1}w_b^{-1})=\prod_{t\in\Omega}(1-w_t^{-2})\cdot\prod_{\substack{a<b\\ a,b\in\Omega}}(1-w_a^{-1}w_b^{-1}).$$

The first factor is $\prod_{t\in\Omega}(c_t^C)^{-1}$; the second pairs with (δ) to give the first product of (1). **This is the only step in the whole derivation at which the type enters, and it enters as the diagonal $a=b$ of a pair product.**

(5) Split $\Phi_{\mathbf{K},\mathbf{K}}$ using $\mathbf{K}^{\mathrm{c}}=\Omega\sqcup\mathbf{M}^{\mathrm{c}}$ and $\Phi_{\mathbf{M},\mathbf{M}}$ using $\mathbf{M}=\mathbf{K}\sqcup\Omega$:

$$\Phi_{\mathbf{K},\mathbf{K}}=\underbrace{\prod_{\substack{k\in\mathbf{K},t\in\Omega\\ t<k}}(1-w_k^{-1}w_t)\prod_{\substack{k\in\mathbf{K},t\in\Omega\\ k<t}}(1-qw_k^{-1}w_t)}_{(\eta)}\cdot(\theta),$$

$$\Phi_{\mathbf{M},\mathbf{M}}=(\theta)\cdot\underbrace{\prod_{\substack{t\in\Omega,j\in\mathbf{M}^{\mathrm{c}}\\ j<t}}(1-w_t^{-1}w_j)\prod_{\substack{t\in\Omega,j\in\mathbf{M}^{\mathrm{c}}\\ t<j}}(1-qw_t^{-1}w_j)}_{(\iota)},$$

where $(\theta)=\prod_{k\in\mathbf{K},j\in\mathbf{M}^{\mathrm{c}},j<k}(1-w_k^{-1}w_j)\prod_{k\in\mathbf{K},j\in\mathbf{M}^{\mathrm{c}},k<j}(1-qw_k^{-1}w_j)$ is *literally the same* product in both lines — in $\Phi_{\mathbf{K},\mathbf{K}}$ it is the part with $u\in\mathbf{M}^{\mathrm{c}}$, in $\Phi_{\mathbf{M},\mathbf{M}}$ the part with $a\in\mathbf{K}$, and both read "$k\in\mathbf{K}$ against $j\in\mathbf{M}^{\mathrm{c}}$, split at k". It cancels, leaving $\Phi_{\mathbf{K},\mathbf{K}}/\Phi_{\mathbf{M},\mathbf{M}}=(\eta)/(\iota)$.

(6) Combine (η) with the surviving $\prod_{k\in\mathbf{K},t\in\Omega}(1-w_k^{-1}w_t)^{-1}$ of step (3). The latter runs over *all* pairs $(k,t)\in\mathbf{K}\times\Omega$; ($\eta$) supplies $(1-w_k^{-1}w_t)$ when $t<k$ and $(1-qw_k^{-1}w_t)$ when $k<t$. Hence the pairs with $t<k$ cancel outright and

$$(\eta)\prod_{\substack{k\in\mathbf{K}\\ t\in\Omega}}\frac{1}{1-w_k^{-1}w_t}=\prod_{\substack{k\in\mathbf{K},t\in\Omega\\ k<t}}\frac{1-qw_k^{-1}w_t}{1-w_k^{-1}w_t},$$

the second product of (1).

(7) Combine (ζ), which runs over all $(t,j)\in\Omega\times\mathbf{M}^{\mathrm{c}}$, with $(\iota)^{-1}$. The pairs with $t<j$ cancel outright and

$$\frac{(\zeta)}{(\iota)}=\prod_{\substack{t\in\Omega,j\in\mathbf{M}^{\mathrm{c}}\\ j<t}}\frac{1-qw_t^{-1}w_j}{1-w_t^{-1}w_j},$$

the third product of (1). Together with the prefactor $(1-q)^{|\Omega|}$ this is $\big(\prod_{t\in\Omega}c_t^C\big)G_{\mathbf{M},\mathbf{K}}$, as claimed.

For type $X=B$ the only changes are in steps (1), (2) and (4). Here

$$\mathrm{Br}^B_{\mathbf{M},\mathbf{K}}=(1-q)^{|\Omega|}\underbrace{\prod_{j\in\mathbf{M}^{\mathrm{c}}}(1-qw_j)\prod_{\substack{a<b\\ a,b\in\mathbf{M}^{\mathrm{c}}}}(1-qw_aw_b)}_{(\alpha')}\prod_{k\in\mathbf{K}}(1-w_k^{-1})\prod_{\substack{a<b\\ a,b\in\mathbf{K}}}(1-w_a^{-1}w_b^{-1})\cdot(\gamma)(\delta)(\zeta)$$

and $D^B_{|\mathbf{M}|}(\mathbf{M})=\prod_{a\in\mathbf{M}}(1-w_a^{-1})\prod_{a<b,\,a,b\in\mathbf{M}}(1-w_a^{-1}w_b^{-1})\cdot(\alpha')$. Step (1) cancels ($\alpha'$). Step (2) uses the *strict* splitting $\prod_{a<b,\,a,b\in\mathbf{M}}=\prod_{a<b,\,a,b\in\mathbf{K}}\prod_{a<b,\,a,b\in\Omega}\prod_{k\in\mathbf{K},t\in\Omega}$, which leaves no diagonal

at all. Step (4) is replaced by the single-index quotient $\prod_{k\in\mathbf{K}}(1-w_k^{-1})/\prod_{a\in\mathbf{M}}(1-w_a^{-1}) = \prod_{t\in\Omega}(1-w_t^{-1})^{-1} = \prod_{t\in\Omega} c_t^B$. Steps (3), (5), (6), (7) involve only $(\gamma),(\delta),(\zeta),\Phi$, which are type-free, and are unchanged. □

Remark 4 (where the whole B/C difference lives, and why it is $w\mapsto w^2$). In the proof the type is consulted exactly once, at step (4) versus its type-B replacement. Type C uses non-strict products $\prod_{a\le b}$ over $\mathbf{K}$ and $\mathbf{M}$, so it carries a diagonal $\prod_a(1-w_a^{-2})$; type B uses strict products plus an explicit single-index $\prod_a(1-w_a^{-1})$. After the $\mathbf{M}$-level normalisation both collapse to a product over Ω, and

$$\frac{c_t^B}{c_t^C} = \frac{1-w_t^{-2}}{1-w_t^{-1}} = 1+w_t^{-1}.$$

So the B/C difference is not merely "single-index" (which is what [C20] measured): it is the substitution $w\mapsto w^2$ applied to one slot, and that slot is the *diagonal of a two-index product*. This is the mechanism Q534/Q538 were asking for, and it is consistent with a boundary rather than a bulk parameter (Q540): a diagonal term $a=b$ of a pair product is exactly a one-variable quantity masquerading as a two-variable one.

Corollary 5. $\widehat{\mathrm{Br}}^X = B^{c^X}$ *in the sense of Definition 2, with* c^B, c^C *as above. Both are free of* q*, hence inversion-invariant, hence Theorem A of [C20] applies to both. Proposition* prop:types *of [C20], previously* `computed`*, is therefore* `proved`*.*

3 The accounting is a bijection — in two different senses

Fix $\mathbf{L}\subsetneq\mathbf{M}\subseteq[N]$ and put $\mathcal{D}:=\mathbf{M}\setminus\mathbf{L}=\{d_1<\cdots<d_m\}$, $m\ge 1$, $x_j:=w_{d_j}$. Then

$$[N] = \mathbf{L}\sqcup\mathcal{D}\sqcup\mathbf{M}^{\mathrm{c}} \tag{4}$$

is a partition. For $\mathbf{U}\subseteq\mathcal{D}$ put $\mathbf{K}(\mathbf{U}):=\mathbf{L}\sqcup\mathbf{U}$; the map $\mathbf{U}\mapsto\mathbf{K}(\mathbf{U})$ is a bijection onto $\{\mathbf{K}:\mathbf{L}\subseteq\mathbf{K}\subseteq\mathbf{M}\}$, and $\epsilon=\epsilon(\mathbf{U})\in\{0,1\}^m$ is defined by $\epsilon_j=[d_j\in\mathbf{U}]$. Set

$$S(\mathbf{U}) := G_{\mathbf{M},\mathbf{K}(\mathbf{U})}(\mathbf{w},q)\; q^{\sum_{d\in\mathbf{U}} d}\; G_{\mathbf{L}^{\mathrm{c}},\mathbf{K}(\mathbf{U})^{\mathrm{c}}}(\mathbf{w},q^{-1}). \tag{5}$$

Substituting $\mathbf{M}\setminus\mathbf{K}(\mathbf{U})=\mathcal{D}\setminus\mathbf{U}$ into (1), the first factor of (5) is $(1-q)^{m-|\mathbf{U}|}$ times the three products

$$\text{(1A)}\ \prod_{\substack{a<b\\ a,b\in\mathcal{D}\setminus\mathbf{U}}},\quad \text{(1B)}\ \prod_{\substack{k\in\mathbf{L}\sqcup\mathbf{U},\ t\in\mathcal{D}\setminus\mathbf{U}\\ k<t}},\quad \text{(1C)}\ \prod_{\substack{t\in\mathcal{D}\setminus\mathbf{U},\ j\in\mathbf{M}^{\mathrm{c}}\\ j<t}} \tag{6}$$

(this is `eq:allcard-SU-first-expanded`, L5093, with the type factor removed). For the second factor, taking complements in $[N]$ and using $\mathbf{A}^{\mathrm{c}}\setminus\mathbf{B}^{\mathrm{c}}=\mathbf{B}\setminus\mathbf{A}$,

$$\mathbf{L}^{\mathrm{c}}\setminus\mathbf{K}(\mathbf{U})^{\mathrm{c}}=\mathbf{U},\qquad \mathbf{K}(\mathbf{U})^{\mathrm{c}}=\mathbf{M}^{\mathrm{c}}\sqcup(\mathcal{D}\setminus\mathbf{U}),\qquad (\mathbf{L}^{\mathrm{c}})^{\mathrm{c}}=\mathbf{L},$$

so that in (1) the roles of $(\mathbf{M},\mathbf{K},\Omega,\mathbf{M}^{\mathrm{c}})$ are played by $(\mathbf{L}^{\mathrm{c}},\mathbf{K}(\mathbf{U})^{\mathrm{c}},\mathbf{U},\mathbf{L})$ and the second factor is $(1-q^{-1})^{|\mathbf{U}|}$ times

$$\text{(2A)}\ \prod_{\substack{a<b\\ a,b\in\mathbf{U}}},\quad \text{(2B)}\ \prod_{\substack{k\in\mathbf{M}^{\mathrm{c}}\sqcup(\mathcal{D}\setminus\mathbf{U}),\ t\in\mathbf{U}\\ k<t}},\quad \text{(2C)}\ \prod_{\substack{t\in\mathbf{U},\ b\in\mathbf{L}\\ b<t}} \tag{7}$$

(`eq:allcard-SU-second-expanded`, L5126), with q replaced by q^{-1} throughout. Each of the six products contributes one *instance* per index pair in its range; let $\mathcal{F}(\mathbf{U})$ denote the multiset of these instances, each tagged by its product of origin and by its unordered index pair.

The two families of consumers are, for $1 \le j \le m$ and $1 \le i < j \le m$,

$$E_j^{(0)} := \prod_{\substack{k\in\mathbf{L}\\ k<d_j}} \frac{1-qw_k^{-1}x_j}{1-w_k^{-1}x_j} \prod_{\substack{b\in\mathbf{M}^{\mathrm{c}}\\ b<d_j}} \frac{1-qx_j^{-1}w_b}{1-x_j^{-1}w_b}, \quad E_j^{(1)} := \prod_{\substack{k\in\mathbf{M}^{\mathrm{c}}\\ k<d_j}} \frac{1-q^{-1}w_k^{-1}x_j}{1-w_k^{-1}x_j} \prod_{\substack{b\in\mathbf{L}\\ b<d_j}} \frac{1-q^{-1}x_j^{-1}w_b}{1-x_j^{-1}w_b}, \tag{8}$$

(`eq:allcard-E0-def` L5161, `eq:allcard-E1-def` L5184) and

$$\mathsf{P}_{00} = \frac{xy-q}{xy-1}, \quad \mathsf{P}_{11} = \frac{xy-q^{-1}}{xy-1}, \quad \mathsf{P}_{10} = \frac{x-qy}{x-y}, \quad \mathsf{P}_{01} = \frac{x-q^{-1}y}{x-y} \tag{9}$$

evaluated at $(x,y) = (x_i, x_j)$.

Lemma 6 (the accounting)**.** *Let $\Phi : \mathcal{F}(\mathbf{U}) \to \mathcal{C}$ be defined on an instance with unordered index pair $\{a,b\}$ by*

$$\Phi(\textit{instance}) = \begin{cases} (i,j) & \text{if } a = d_i,\ b = d_j \text{ with } i<j, \\ j & \text{if exactly one of } a,b \text{ equals } d_j, \end{cases}$$

with codomain $\mathcal{C} := \big\{(a,j) : a \in \mathbf{L}\cup\mathbf{M}^{\mathrm{c}},\ 1\le j\le m,\ a<d_j\big\} \sqcup \big\{(i,j) : 1\le i<j\le m\big\}$, the first component of the mixed case being the index not in $\mathcal{D}$. Then:

(i) *Φ is well defined: no instance has both indices outside $\mathcal{D}$.*

(ii) *Φ is surjective.*

(iii) *Φ is injective.*

(iv) *The instance over (a,j) has the value of the corresponding factor of $E_j^{(\epsilon_j)}$ in (8); the instance over (i,j) has the value $\mathsf{P}_{\epsilon_i\epsilon_j}(x_i,x_j)$ of (9).*

Consequently

$$S(\mathbf{U}) = q^{\sum_j d_j\epsilon_j}(1-q)^{m-|\mathbf{U}|}(1-q^{-1})^{|\mathbf{U}|} \prod_{j=1}^{m} E_j^{(\epsilon_j)} \prod_{i<j} \mathsf{P}_{\epsilon_i\epsilon_j}(x_i,x_j). \tag{10}$$

Proof. **(i).** Each of the six products has a distinguished index constrained to $\mathcal{D}$: in (1A) and (2A) both indices lie in $\mathcal{D}\setminus\mathbf{U}$ resp. $\mathbf{U}$; in (1B) and (1C) the index t lies in $\mathcal{D}\setminus\mathbf{U}$; in (2B) and (2C) the index t lies in $\mathbf{U}$. Since $\mathbf{U}\subseteq\mathcal{D}$ and $\mathcal{D}\setminus\mathbf{U}\subseteq\mathcal{D}$, every instance has at least one index in $\mathcal{D}$.

(ii), mixed pairs. Fix $a \in \mathbf{L}\cup\mathbf{M}^{\mathrm{c}}$ and j with $a<d_j$. By (4), a lies in exactly one of $\mathbf{L}$, $\mathbf{M}^{\mathrm{c}}$; and d_j lies in exactly one of $\mathbf{U}$, $\mathcal{D}\setminus\mathbf{U}$. The four combinations are realised as follows, and in each case the order hypothesis of the relevant product is *satisfied* because it reads precisely $a<d_j$:

$$\begin{array}{lll} a\in\mathbf{L}, & \epsilon_j=0: & \text{(1B) with } k=a,\ t=d_j \quad (k\in\mathbf{L}\subseteq\mathbf{L}\sqcup\mathbf{U},\ t\in\mathcal{D}\setminus\mathbf{U}), \\ a\in\mathbf{L}, & \epsilon_j=1: & \text{(2C) with } b=a,\ t=d_j, \\ a\in\mathbf{M}^{\mathrm{c}}, & \epsilon_j=0: & \text{(1C) with } j=a,\ t=d_j, \\ a\in\mathbf{M}^{\mathrm{c}}, & \epsilon_j=1: & \text{(2B) with } k=a,\ t=d_j \quad (k\in\mathbf{M}^{\mathrm{c}}\subseteq\mathbf{M}^{\mathrm{c}}\sqcup(\mathcal{D}\setminus\mathbf{U})). \end{array}$$

(ii), $\mathcal{D}$-pairs. Fix $i<j$, so $d_i<d_j$. The four values of (ϵ_i,ϵ_j) are realised by

$$(0,0):\ (1\text{A}),\quad (1,1):\ (2\text{A}),\quad (1,0):\ (1\text{B})\text{ with } k=d_i\in\mathbf{U},\ t=d_j,\quad (0,1):\ (2\text{B})\text{ with } k=d_i,\ t=d_j\in\mathbf{U},$$

the last two again having $k<t$ satisfied because $d_i<d_j$.

(iii), mixed pairs. Suppose an instance has indices $a\notin\mathcal{D}$ and $d_j\in\mathcal{D}$. It cannot come from (1A) or (2A), whose both indices lie in $\mathcal{D}$. In (1B) the index t is in $\mathcal{D}$ and k ranges over $\mathbf{L}\sqcup\mathbf{U}$; since $a\notin\mathcal{D}$ we need $k=a\in\mathbf{L}$ and $t=d_j\in\mathcal{D}\setminus\mathbf{U}$, i.e. $\epsilon_j=0$, and the product requires $a<d_j$. In (2C), $t\in\mathbf{U}$ and $b\in\mathbf{L}$, forcing $b=a\in\mathbf{L}$, $\epsilon_j=1$, $a<d_j$. In (1C), $t\in\mathcal{D}\setminus\mathbf{U}$ and $j\in\mathbf{M}^{\mathrm{c}}$, forcing $a\in\mathbf{M}^{\mathrm{c}}$, $\epsilon_j=0$, $a<d_j$. In (2B), $t\in\mathbf{U}$ and $k\in\mathbf{M}^{\mathrm{c}}\sqcup(\mathcal{D}\setminus\mathbf{U})$, forcing $k=a\in\mathbf{M}^{\mathrm{c}}$, $\epsilon_j=1$, $a<d_j$. So at most one of the four fires: the pair ($\mathbf{L}$ or $\mathbf{M}^{\mathrm{c}}$) is decided by (4) — these sets are disjoint — and the pair ($\epsilon_j=0$ or 1) is decided by $\mathbf{U}$. Both discriminators are *partitions of index sets*; the order relation plays no part.

(iii), $\mathcal{D}$-pairs. Let both indices be in $\mathcal{D}$, say d_i,d_j with $i<j$. If $\epsilon_i=\epsilon_j=0$ then only (1A) can fire: (1B) and (2B) need an index in $\mathbf{U}$ or in $\mathbf{L}\cup\mathbf{M}^{\mathrm{c}}$ on one side, (1C) and (2C) need an index outside $\mathcal{D}$, and (2A) needs both in $\mathbf{U}$. Dually for $\epsilon_i=\epsilon_j=1$, only (2A). Now suppose $\epsilon_i\neq\epsilon_j$. Then *two* products have the pair in range, and they are separated only by the order condition:

- $(\epsilon_i,\epsilon_j)=(1,0)$. In (1B) one may take $k=d_i\in\mathbf{U}$, $t=d_j\in\mathcal{D}\setminus\mathbf{U}$, and $k<t$ holds. In (2B) the same pair would need $t\in\mathbf{U}$, hence $t=d_i$, and $k\in\mathcal{D}\setminus\mathbf{U}$, hence $k=d_j$; but then $k<t$ reads $d_j<d_i$, which is false. So (2B) does not fire.
- $(\epsilon_i,\epsilon_j)=(0,1)$. In (2B) one may take $k=d_i\in\mathcal{D}\setminus\mathbf{U}$, $t=d_j\in\mathbf{U}$, and $k<t$ holds. In (1B) the same pair would need $t\in\mathcal{D}\setminus\mathbf{U}$, hence $t=d_i$, and $k\in\mathbf{U}$, hence $k=d_j$; but then $k<t$ reads $d_j<d_i$, false. So (1B) does not fire.

In both cases the surviving product is the one in which the *smaller* index d_i sits in the "$\mathbf{K}$" slot and the larger d_j in the "Ω" slot. Hence exactly one instance.

(iv). Read off the factors. For a mixed pair with $a\in\mathbf{L}$, $\epsilon_j=0$: (1B) contributes $\frac{1-qw_a^{-1}w_{d_j}}{1-w_a^{-1}w_{d_j}}=\frac{1-qw_a^{-1}x_j}{1-w_a^{-1}x_j}$, the first product of $E_j^{(0)}$. For $a\in\mathbf{M}^{\mathrm{c}}$, $\epsilon_j=0$: (1C) with $t=d_j$ contributes $\frac{1-qw_{d_j}^{-1}w_a}{1-w_{d_j}^{-1}w_a}=\frac{1-qx_j^{-1}w_a}{1-x_j^{-1}w_a}$, the second product of $E_j^{(0)}$. For $a\in\mathbf{M}^{\mathrm{c}}$, $\epsilon_j=1$: (2B) at q^{-1} contributes $\frac{1-q^{-1}w_a^{-1}x_j}{1-w_a^{-1}x_j}$, the first product of $E_j^{(1)}$; for $a\in\mathbf{L}$, $\epsilon_j=1$: (2C) at q^{-1} contributes $\frac{1-q^{-1}x_j^{-1}w_a}{1-x_j^{-1}w_a}$, the second. For $\mathcal{D}$-pairs, with $x=x_i$, $y=x_j$:

$$(1\text{A}):\frac{1-q(xy)^{-1}}{1-(xy)^{-1}}=\mathsf{P}_{00},\quad (2\text{A}):\frac{1-q^{-1}(xy)^{-1}}{1-(xy)^{-1}}=\mathsf{P}_{11},\quad (1\text{B}):\frac{1-qx^{-1}y}{1-x^{-1}y}=\mathsf{P}_{10},\quad (2\text{B}):\frac{1-q^{-1}x^{-1}y}{1-x^{-1}y}=\mathsf{P}_{01}.$$

Finally (10): the prefactors of the two factors of (5) are $(1-q)^{m-|\mathbf{U}|}$ and $(1-q^{-1})^{|\mathbf{U}|}$, the explicit power of q in (5) is $q^{\sum_{d\in\mathbf{U}}d}=q^{\sum_j d_j\epsilon_j}$, and by (i)–(iv) the remaining factors are exactly $\prod_j E_j^{(\epsilon_j)}\prod_{i<j}\mathsf{P}_{\epsilon_i\epsilon_j}$, each occurring once. (1) has no other factors. □

Remark 7 (P1 came back SPLIT: "bijection" is two claims). I pre-registered P1 as "the map pair → consumer is a bijection, and the proof needs no hypothesis beyond $\mathbf{L}\subsetneq\mathbf{M}$". The conclusion holds and the hypothesis claim holds — nothing beyond (4) and $\mathbf{U}\subseteq\mathcal{D}$ is used. But the two halves of "bijection" do not have one answer, and the grouping asserted that they did:

- **Surjectivity** (nothing dropped) uses only the set partition (4) and $\mathcal{D} = \mathbf{U} \sqcup (\mathcal{D} \setminus \mathbf{U})$. Every order condition it meets is *satisfied, never needed*: in (ii) the condition $k < t$ reads $a < d_j$ or $d_i < d_j$, both of which hold by hypothesis.
- **Injectivity** (nothing double-counted) splits again. On mixed pairs it is also pure partition logic: $\mathbf{L} \cap \mathbf{M}^{\mathrm{c}} = \varnothing$ and $\mathbf{U} \cap (\mathcal{D} \setminus \mathbf{U}) = \varnothing$ do all the work. But on $\mathcal{D}$-pairs with $\epsilon_i \neq \epsilon_j$ *two* products have the pair in range, and the only thing that separates them is the *total order* on $[N]$. Delete "$k < t$" and the accounting fails — by over-counting, not by dropping.

So the strict order condition is **irremovable from injectivity and inert for surjectivity**, and the same condition is what forces $\mathsf{P}_{01} \neq \mathsf{P}_{10}$: the asymmetry of the two off-diagonal shapes is the shadow of an order relation, not of anything about $\mathbf{L}$ or $\mathbf{M}$. Two pre-registered lesions confirm the split by breaking *disjoint* halves (§7, L1 and L5).

4 The three ratios and the q-exponent

Lemma 8. *With the notation of §3:*

(a) $E_j^{(1)} = q^{-(d_j - j)} E_j^{(0)}$ *for* $1 \le j \le m$.

(b) *With* $H_{00} = xy - q$, $H_{11} = qxy - 1$, $H_{01} = \frac{(qx-y)(xy-1)}{x-y}$, $H_{10} = \frac{(x-qy)(xy-1)}{x-y}$ *one has* $\mathsf{P}_{ab}(x,y) = q^{-b} H_{ab}(x,y)/(xy-1)$ *for all* $a, b \in \{0,1\}$*; in particular* $\mathsf{P}_{ab}/\mathsf{P}_{00} = q^{-b} H_{ab}/H_{00}$.

(c) $S(\mathbf{U})/S(\varnothing) = (-1)^{|\mathbf{U}|} \prod_{i<j} H_{\epsilon_i \epsilon_j}(x_i, x_j)/H_{00}(x_i, x_j)$.

Proof. **(a).** Put $z = w_a/x_j$ for $a \in \mathbf{L}$ with $a < d_j$. The factor of $E_j^{(1)}$ at a is $\frac{1-q^{-1}z}{1-z} = \frac{q-z}{q(1-z)}$, while the factor of $E_j^{(0)}$ at a is $\frac{1-q/z}{1-1/z} = \frac{z-q}{z-1} = \frac{q-z}{1-z}$; their ratio is q^{-1}. For $a \in \mathbf{M}^{\mathrm{c}}$ with $a < d_j$, put $z = x_j/w_a$; the factor of $E_j^{(1)}$ is $\frac{1-q^{-1}z}{1-z}$ and that of $E_j^{(0)}$ is $\frac{1-q/z}{1-1/z}$, the same two expressions, ratio q^{-1} again. By Lemma 6(iv) both $E_j^{(0)}$ and $E_j^{(1)}$ are indexed by *the same* set of a, namely $\{a \in \mathbf{L} \cup \mathbf{M}^{\mathrm{c}} : a < d_j\}$ — only the shape of each factor changes — so $E_j^{(1)}/E_j^{(0)} = q^{-\#\{a \notin \mathcal{D} : a < d_j\}}$. Of the $d_j - 1$ indices below d_j, exactly $j-1$ lie in $\mathcal{D}$ (namely $d_1, \dots, d_{j-1}$, since $\mathcal{D}$ is listed in increasing order), and by (4) all the others lie in $\mathbf{L} \cup \mathbf{M}^{\mathrm{c}}$. Hence the exponent is $(d_j - 1) - (j-1) = d_j - j$.

(b). Four checks. $q^0 H_{00}/(xy-1) = \frac{xy-q}{xy-1} = \mathsf{P}_{00}$. $q^{-1} H_{11}/(xy-1) = \frac{qxy-1}{q(xy-1)} = \frac{xy-q^{-1}}{xy-1} = \mathsf{P}_{11}$. $q^0 H_{10}/(xy-1) = \frac{(x-qy)(xy-1)}{(x-y)(xy-1)} = \frac{x-qy}{x-y} = \mathsf{P}_{10}$. $q^{-1} H_{01}/(xy-1) = \frac{(qx-y)(xy-1)}{q(x-y)(xy-1)} = \frac{x-q^{-1}y}{x-y} = \mathsf{P}_{01}$. (Motegi says "it is easy to check"; the four lines are the check.)

(c). Divide (10) by its value at $\mathbf{U} = \varnothing$, where all $\epsilon_j = 0$. The prefactors give

$$q^{\sum_j d_j \epsilon_j} \Big(\frac{1 - q^{-1}}{1 - q} \Big)^{|\mathbf{U}|} = q^{\sum_j d_j \epsilon_j} (-q^{-1})^{|\mathbf{U}|} = (-1)^{|\mathbf{U}|} q^{\sum_j (d_j - 1)\epsilon_j}.$$

By (a), $\prod_j E_j^{(\epsilon_j)}/E_j^{(0)} = q^{-\sum_j (d_j - j)\epsilon_j}$. By (b), and since for each j there are exactly $j-1$ indices $i < j$,

$$\prod_{i<j} \frac{\mathsf{P}_{\epsilon_i \epsilon_j}}{\mathsf{P}_{00}} = q^{-\sum_{i<j} \epsilon_j} \prod_{i<j} \frac{H_{\epsilon_i \epsilon_j}}{H_{00}} = q^{-\sum_j (j-1)\epsilon_j} \prod_{i<j} \frac{H_{\epsilon_i \epsilon_j}}{H_{00}}.$$

The total exponent of q is $\sum_j \epsilon_j \big[(d_j - 1) - (d_j - j) - (j-1) \big] = 0$, which gives (c). □

Remark 9. The three cancelling contributions (d_j-1), $-(d_j-j)$, $-(j-1)$ come from three unrelated places — the prefactor $(1-q^{-1})/(1-q)$, a count of indices *outside* $\mathcal{D}$ below d_j, and a count of indices *inside* $\mathcal{D}$ below d_j — and the last two are complementary by (4). The identity $(d_j-1)=(d_j-j)+(j-1)$ is that complementarity and nothing more.

5 The cancellation lemma

Lemma 10 (Motegi, `lem:allcard-universal-cancellation`, L4790)**.** *For $m\geq 1$ and indeterminates $x_1,\dots,x_m$, in $\mathbb{Q}(q,x_1,\dots,x_m)$,*

$$F_m(x_1,\dots,x_m):=\sum_{\epsilon\in\{0,1\}^m}(-1)^{m-|\epsilon|}\prod_{1\leq i<j\leq m}H_{\epsilon_i\epsilon_j}(x_i,x_j)=0. \tag{11}$$

This is Motegi's lemma with Motegi's proof. I reproduce it because the present note must be self-contained, and because three of its steps are left to the reader; I have checked every step and the three gaps are filled below. One checks directly from (9)–(11) that $H_{ab}(x,y)=H_{ba}(y,x)$ for all a,b.

Proof. Regard F_m as a rational function of x_m over $\mathbb{Q}(q,x_1,\dots,x_{m-1})$.

No poles. The only denominators in (11) are the factors x_i-x_j occurring in H_{01},H_{10}, so the only possible poles in x_m are at $x_m=x_i$, $i<m$, and in each summand only the single factor $H_{\epsilon_i\epsilon_m}(x_i,x_m)$ has that pole, which is simple. Fix $i<m$ and fix ϵ_k for $k\neq i,m$. Terms with $\epsilon_i=\epsilon_m$ have no pole at $x_m=x_i$. The two terms with $(\epsilon_i,\epsilon_m)=(0,1)$ and $(1,0)$ carry the same sign $(-1)^{m-|\epsilon|}$, since $|\epsilon|$ is the same. *First filled gap:* the remaining factors agree, because for $k\neq i,m$ the product of the two factors involving x_k and one of x_i,x_m is, at $x_m=x_i$ and using $H_{ab}(x,y)=H_{ba}(y,x)$ to put both in the same orientation, $H_{\epsilon_k\epsilon_i}(x_k,x_i)\,H_{\epsilon_k\epsilon_m}(x_k,x_i)$ when $k<i$ and $H_{\epsilon_i\epsilon_k}(x_i,x_k)\,H_{\epsilon_m\epsilon_k}(x_i,x_k)$ when $k>i$; either expression is symmetric under $\epsilon_i\leftrightarrow\epsilon_m$. Factors involving neither i nor m are untouched. Since

$$\operatorname*{Res}_{y=x}H_{01}(x,y)=x(1-q)(x^2-1)=-\operatorname*{Res}_{y=x}H_{10}(x,y),$$

the two residues cancel. Hence F_m is a polynomial in x_m.

Degree. Second filled gap: dividing out, $H_{ab}(x,y)=q^a xy+O(1)$ as $y\to\infty$ for each of the four (a,b) — e.g. $H_{10}(x,y)=qxy-\big((1-q)x^2+q\big)+\frac{(q-1)x^3+\dots}{x-y}$ — and in particular the leading coefficient $q^a x$ does not depend on the second index b. Fix $\epsilon_1,\dots,\epsilon_{m-1}$; the factors of the summand containing x_m are $\prod_{i<m}H_{\epsilon_i\epsilon_m}(x_i,x_m)=q^{\epsilon_1+\cdots+\epsilon_{m-1}}\big(\prod_{i<m}x_i\big)x_m^{m-1}+O(x_m^{m-2})$, whose leading term is the same for $\epsilon_m=0$ and $\epsilon_m=1$, while the two signs $(-1)^{m-|\epsilon|}$ are opposite. So the coefficients of x_m^{m-1} cancel in pairs and $\deg_{x_m}F_m\leq m-2$.

Roots. For $m=1$, $F_1=-1+1=0$. For $m=2$, $\deg_{x_2}F_2\leq 0$ so F_2 does not involve x_2; and $F_2(x_1,x_1^{-1})=H_{00}(x_1,x_1^{-1})+H_{11}(x_1,x_1^{-1})=(1-q)+(q-1)=0$, because H_{01} and H_{10} both carry the factor $xy-1$ and vanish at $y=x^{-1}$. Hence $F_2=0$. Let $m\geq 3$ and assume (11) for $m-2$. Fix $k<m$ and specialise $x_m=x_k^{-1}$. Since $H_{01}(x_k,x_k^{-1})=H_{10}(x_k,x_k^{-1})=0$, only the ϵ with $\epsilon_k=\epsilon_m=:s$ survive, and the surviving (k,m) factor is $H_{ss}(x_k,x_k^{-1})=(-1)^s(1-q)$. *Third filled gap:* for $j\neq k,m$ the two factors involving x_j and one of x_k,x_m become, after orienting with $H_{ab}(x,y)=H_{ba}(y,x)$, $H_{\eta,s}(u,x_k)H_{\eta,s}(u,x_k^{-1})$ with $u=x_j$, $\eta=\epsilon_j$; and

$$H_{\eta,0}(u,x)H_{\eta,0}(u,x^{-1})=\frac{(ux-q)(u-qx)}{x},\qquad H_{\eta,1}(u,x)H_{\eta,1}(u,x^{-1})=\frac{(qu-x)(qux-1)}{x}$$

for *both* $\eta = 0$ and $\eta = 1$ — the four products are computed using $u - qx^{-1} = (ux - q)/x$, $ux^{-1} - 1 = (u - x)/x$, $u - x^{-1} = (ux - 1)/x$, which make the $(u - x)$ and $(ux - 1)$ factors cancel pairwise. So all factors involving k or m are independent of the remaining bits and come outside the alternating sum over them. The remaining sum is $\sum_{\epsilon'} (-1)^{m-|\epsilon'|-2s} \prod H_{\epsilon'_i \epsilon'_j} = \sum_{\epsilon'} (-1)^{(m-2)-|\epsilon'|} \prod H_{\epsilon'_i \epsilon'_j} = F_{m-2}(x_1, \ldots, \widehat{x_k}, \ldots, x_{m-1}) = 0$ by induction. Therefore $F_m|_{x_m = x_k^{-1}} = 0$ for $k = 1, \ldots, m-1$. A polynomial in x_m of degree $\leq m-2$ over the field $\mathbb{Q}(q, x_1, \ldots, x_{m-1})$ with the $m-1$ distinct roots $x_1^{-1}, \ldots, x_{m-1}^{-1}$ is zero. □

6 The theorem

Theorem 11. *For all* $\mathbf{M}, \mathbf{L} \subseteq [N]$, $T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}} = \delta_{\mathbf{M},\mathbf{L}}$.

Proof. Support. $G_{\mathbf{M},\mathbf{K}} = 0$ unless $\mathbf{K} \subseteq \mathbf{M}$, and $G_{\mathbf{L}^c,\mathbf{K}^c} = 0$ unless $\mathbf{K}^c \subseteq \mathbf{L}^c$, i.e. unless $\mathbf{L} \subseteq \mathbf{K}$. So every nonzero summand of (2) has $\mathbf{L} \subseteq \mathbf{K} \subseteq \mathbf{M}$, and $T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}} = 0$ unless $\mathbf{L} \subseteq \mathbf{M}$.

Diagonal. If $\mathbf{L} = \mathbf{M}$ then $\mathbf{K} = \mathbf{M}$ is forced; $\Omega = \varnothing$ makes every product in (1) empty, so $G_{\mathbf{A},\mathbf{A}} = 1$ for every $\mathbf{A}$, and the unique summand is $1 \cdot q^0 \cdot 1 = 1$.

Off-diagonal. Let $\mathbf{L} \subsetneq \mathbf{M}$ and adopt the notation of §3. Then $T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}} = \sum_{\mathbf{U} \subseteq \mathcal{D}} S(\mathbf{U})$. By Lemma 8(c),

$$\sum_{\mathbf{U} \subseteq \mathcal{D}} S(\mathbf{U}) = \frac{S(\varnothing)}{\prod_{i<j} H_{00}(x_i, x_j)} \sum_{\epsilon \in \{0,1\}^m} (-1)^{|\epsilon|} \prod_{i<j} H_{\epsilon_i \epsilon_j}(x_i, x_j),$$

using that $\mathbf{U} \mapsto \epsilon(\mathbf{U})$ is a bijection $2^{\mathcal{D}} \to \{0,1\}^m$ with $|\mathbf{U}| = |\epsilon|$, and that each $H_{00}(x_i, x_j) = x_i x_j - q$ is a nonzero element of F, so the division is legitimate. Since $(-1)^{|\epsilon|} = (-1)^m (-1)^{m-|\epsilon|}$, the sum is $(-1)^m F_m$, which vanishes by Lemma 10. Hence $T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}} = 0$. □

Corollary 12. *`prop:allcard-Bhat-inverse`* (L4897) *holds for* $X \in \{B, C\}$ *and all* $N \geq 1$*:* $\big(\widehat{\mathrm{Br}}^X(\mathbf{w}, q)^{-1}\big)_{\mathbf{I},\mathbf{M}} = q^{|\lambda(\mathbf{I})|-|\lambda(\mathbf{M})|} \widehat{\mathrm{Br}}^X_{\mathbf{M}^c,\mathbf{I}^c}(\mathbf{w}, q^{-1})$.

Proof. By Lemma 3/Corollary 5, $\widehat{\mathrm{Br}}^X = B^{c^X}$ with c^X free of q, hence inversion-invariant. By Theorem A of [C20], $T^{c^X}_{\mathbf{M},\mathbf{L}} = \big(\prod_{d \in \mathbf{M} \setminus \mathbf{L}} c^X_d\big) T^{\mathbf{1}}_{\mathbf{M},\mathbf{L}}$ for $\mathbf{L} \subsetneq \mathbf{M}$ and $T^{c^X} = T^{\mathbf{1}}$ otherwise. By Theorem 11, $T^{\mathbf{1}} = \delta$; each $c^X_d \in F^\times$, so $T^{c^X}_{\mathbf{M},\mathbf{L}} = \delta_{\mathbf{M},\mathbf{L}}$. The entry $T^{c^X}_{\mathbf{M},\mathbf{L}}$ is precisely the $(\mathbf{M}, \mathbf{L})$ entry of $\widehat{\mathrm{Br}}^X(\mathbf{w}, q)$ times the matrix on the right of the display (`eq:allcard-boundary-product`, L4921), so that product is the identity, and the right-hand matrix is the inverse. □

Remark 13 (P2, confirmed with a mechanism). I pre-registered that the accounting is type-independent. It is, and vacuously so: by Lemma 3 the type has already been extracted as $\prod_{t \in \Omega} c_t^X$ before §3 begins, and the type appears at exactly one line of the entire development (step (4) of Lemma 3). The sub-prediction — that type C's c is the *diagonal* of a pair product while type B's is a genuine single-index product — is confirmed by direct reading of `eq:allcard-Br-C` against `eq:allcard-Br-B` and `eq:local-DC-orth` against `eq:local-DB-orth`: type C writes $\prod_{a \leq b}$ where type B writes $\prod_{a<b}$ together with $\prod_a$. See Remark 4.

7 Verification

Pure Python with SymPy; q and $w_1, \ldots, w_N$ are symbols throughout and no numeric value is ever substituted for q (on 2026-10-18 a numeric q colliding with a symbolic one broke exactly the two

of six arms that referenced the shared name). Code in `scratch/g5/`; pre-registration, written before any computation, in `scratch/2026-10-21-prove-prereg.md`. Comparisons of collections are *set/multiset* comparisons in both directions, never comparisons of totals.

Claim	Domain	Result
Lemma 3, both types, from raw Br^X, D^X, Φ	$N \le 4$, all $\mathbf{K} \subseteq \mathbf{M}$, $\mathbf{M} \neq \varnothing$	232/232
Lemma 6(i)–(iii), multiset of instances	$N \le 5$, all 992 triples $(\mathbf{M}, \mathbf{L}, \mathbf{U})$	992/992
Lemma 6, eq. (10) (values)	$N \le 4$, all 310 triples	310/310
Lemma 8(a) $E_j^{(1)} = q^{-(d_j - j)} E_j^{(0)}$	$N \le 5$, all $(\mathbf{M}, \mathbf{L}, j)$	547/547
Lemma 8(b) all four (a, b)	symbolic in x, y, q	4/4
Lemma 10 $F_m = 0$	fully symbolic, $m \le 5$	5/5
Lemma 10 $F_m = 0$	three exact rational x-points, q symbolic, $m \le 7$	21/21

The declared positive control (P3). A green at a vacuous instance carries no information: this month a lemma lost two different hypotheses on consecutive days and both times $r = 0$ satisfied every hypothesis. So the control was *named before computing*:

$$N = 6, \qquad \mathbf{M} = \{2, 3, 5\}, \qquad \mathbf{L} = \{3\}, \qquad \mathcal{D} = \{2, 5\}, \qquad \mathbf{M}^{\mathrm{c}} = \{1, 4, 6\}.$$

It is non-vacuous in the way that matters: $\mathbf{L}$, $\mathcal{D}$, $\mathbf{M}^{\mathrm{c}}$ are all non-empty and interleaved, $m = 2$ so a $\mathcal{D}$-pair exists, both branches of each E_j are non-empty, and — the discriminating property — $(d_j - j) = (1, 3)$ is *different* from $(d_j - 1) = (1, 4)$ at $j = 2$. On this instance (10) is exact for all four $\mathbf{U}$ and $\sum_{\mathbf{U}} S(\mathbf{U}) = 0$.

Lesions. Five were pre-registered; all five break, and the two that break Lemma 6 break *disjoint halves* of it, which is the measurement behind Remark 7.

Lesion	Effect	Verdict
L1: delete $k < t$ from (1B)	110/240 mismatches; witness *extra* $\{(1, 2), \mathsf{P}_{01}\}$, *nothing missing*	breaks injectivity
L5: drop $a < d_j$ from the E_j range	176/240; witness *missing* only, nothing extra	breaks surjectivit
L2: swap $\mathsf{P}_{01} \leftrightarrow \mathsf{P}_{10}$	(10) fails at $\mathbf{U} = \{2\}, \{5\}$ on the control	breaks
L3: $q^{-(d_j - 1)}$ for $q^{-(d_j - j)}$	fails at $\mathbf{U} = \{5\}, \{2, 5\}$, i.e. exactly where $\epsilon_2 = 1$	breaks
L4: delete the sign $(-1)^{m - \lvert\epsilon\rvert}$	unsigned sum nonzero, $m = 2, 3, 4$	breaks

L3 is the reason the control was chosen as it was: it fails only at the $\mathbf{U}$ for which $\epsilon_2 = 1$, and only because $d_2 - 2 \neq d_2 - 1$ there. On a control with $\mathcal{D}$ an initial segment, L3 would have survived.

What the census does *not* show. The identity $T^{\mathbf{1}} = \delta$ itself was already verified 336/336 on 2026-10-19 and 340/340 on 2026-10-20, and that did not close the gap, because a census of the conclusion is not a derivation of the step. The rows above are tests of the *steps*: a multiset of factor instances, a per-summand factorisation, three ratios. Those are what was missing.

8 What is now closed, and what G5 still rests on

Node	Before today	After
`motegi-lem-allcard-Bhat-product` (L4660)	`computed` (240/240)	`proved` (Lemma 3)
`g5-two-index-accounting` (L5263)	not a node	`proved` (Lemma 6)
`motegi-lem-universal-cancellation` (L4790)	audited, not graded	`proved` (Lemma 10)
`g5-core-c-equals-one`	open	`proved` (Theorem 11)
`motegi-prop-allcard-Bhat-inverse` (L4897)	`computed`	`proved` (Cor. 12)

G5 itself is not closed, and saying otherwise would be an over-claim in my own favour. `thm:BC-all-cardinality-biorthogonality` needs three inputs; today settles one of them outright. The other two remain premises, and I name them rather than leave them inside a prose field where no boundary rule can see them:

P1. The cross-sector branching formulas (`eq:typeC-cross-sector-branching` L2920, `eq:typeB-cross-sector` L3685). Everything in this note starts *downstream* of them: `eq:allcard-Br-B/C` are definitions assembled from them, and Lemma 3 verifies $\widehat{\mathrm{Br}}^X$ against those definitions, not the definitions against the branching theorem. This premise is already in my registry as `motegi-branching-formula` at trust `proved` — the formula itself (in its one-summand form `eq:typeC-one-branch-explicit` L3233, which gives L2920 on summing over $\mathbf{K}$) read first-hand together with the final step of its proof, and independently corroborated at $N = 2, 3$ by an engine that evaluates the S_n symmetrisation formula and never touches the branching formula. What is *not* read is the derivation preceding L3203.

But the premise is asymmetric between the types, and that asymmetry is new to me today. Type C has a proof region. Type B does not: at L3598 Motegi writes "*The proofs of the type B statements below are parallel to the corresponding type C arguments, with the type B specialization and normalization factors in place of the type C ones, and are therefore omitted.*" So `eq:typeB-cross-sector-branching` is stated with no proof in the paper at all — declared, not shown. For the present result this matters less than it looks, and in one specific sense today's work *discharges* the parallelism claim where I need it: the type-B/type-C parallelism of the *all-cardinality biorthogonality* is no longer an assertion but a theorem, because Lemma 3 carries out both types explicitly from their raw definitions and Theorem A of [C20] shows the one surviving difference is invisible to the identity. What remains undischarged is strictly the step from $\widetilde{WB}$ to Br^B, i.e. the omitted type-B branching derivation.

Correction to an earlier draft of this page. I first wrote that this premise is `unclassified` and that I "have not read" it. Both are wrong, and wrong in the direction that costs me nothing — an under-claim about my own evidence has no natural objector. The node exists, at `proved`, with its read depth recorded; `eq:typeC-one-branch-explicit` is read first-hand. The accurate statement is the paragraph above.

P2. `motegi-lem-allcard-Phihat-inverse` (L4425), the $\widehat{\Phi}$ half of the inverse. Its chain was traced on 2026-10-20 to two citations of Rimányi–Tarasov–Varchenko — Thm. 6.6/eq. (6.7) and Thms. 3.1/6.7 — whose *statements* I have read first-hand and whose transcription into [Mot] I checked; RTV's *proofs* (their §7.5) I have not read, which is the ordinary standard for a published refereed result. What is *not* checked is Motegi's derivation of L4425 from them. Trust: `published` at the RTV leaves, `unclassified` at the derivation.

So the precise statement of today's result is: *the half of G5 that carries all the type dependence and none of the type-A input is now proved, gap-free, for both types and all N, from Motegi's branching formulas as a black box.* The two `proved` grades that rest on G5 (my `thm:S`, `thm:S'`) still rest on P1 and P2 above, and should be read that way until those are audited too.

9 What remains open

1. **The omitted type-B branch** (L3598) is the cheapest remaining target, and it is more specific than "read the branching derivation". Type C's derivation L2920–L3400 is unread

but corroborated; type B's does not exist on the page. The task is to run Motegi's type-C argument with the type-B specialization and normalization factors substituted, and check that "parallel" is true rather than plausible. Today's Lemma 3 is a proof of concept that it is: at the one place where the types could have diverged (step (4)), they diverge by exactly a diagonal-versus-single-index bookkeeping difference and by nothing else.

2. **P2 above**: Motegi's derivation of `lem:allcard-Phihat-inverse` from RTV, L4425–L4657. This is the only remaining branch with type-A content.

3. **Q540**, sharpened by Remark 4: the type parameter is the diagonal $a = b$ of a pair product. The prediction that it is a *boundary* parameter now has a sharper falsifier — exhibit a two-variable kernel in this family whose *off*-diagonal entries differ between types. Lemma 3 says there is none in $\widehat{\mathrm{Br}}^X$; the question is whether the same holds upstream, in the objects of P1.

4. **G3** (`(BD)` in type B: $\mathrm{ord}_S \tilde{A}_{\mathbf{I},\mathbf{I}} = 1$ for $m \in \mathbf{I}$) was the secondary target and was not reached, for the second day.

References

[Mot] K. Motegi, *Type B- and C- trigonometric weight functions*, `arXiv:2610.15344`.

[C20] Clio, *The row-factor half of G5*, 20 October 2026, `proofs/2026-10-20-c2-G5-reduction.tex`; Theorem A and Proposition `prop:types` therein.

[RTV] R. Rimányi, V. Tarasov, A. Varchenko, *Trigonometric weight functions as K-theoretic stable envelope maps for the cotangent bundle of a flag variety*; local copy `memory/reading/2026-10-18-browse-web-artifacts/rtv-ktheory.pdf`.